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COMPUTER SCIENCE ATAR

Systems Analysis and Development

TYPES OF SYSTEM DEVELOPMENT METHODOLOGIES

LINEAR - WATERFALL/CASCADE

The waterfall/cascade model is a sequential design process in which progress is seen flowing down through the stages. Each phase must be completed before the next phase can begin, hence being easy and simple to understand. A review is conducted at the end of each phase to make sure the project is being carried out correctly.

Advantages

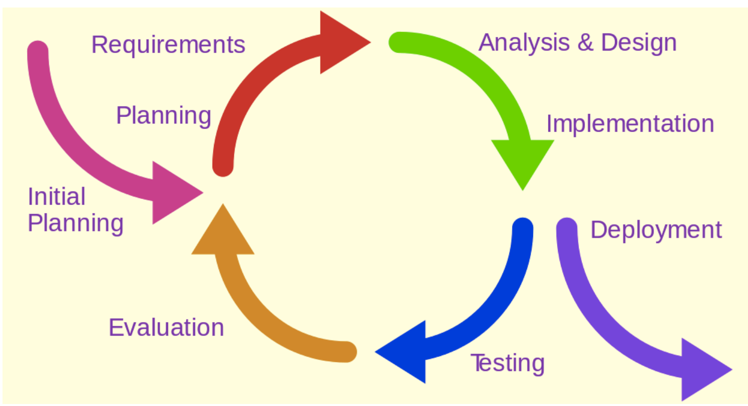
- Spending time and effort at the requirements stage saves time and money overall. The waterfall method encourages this by producing comprehensive documentation for the requirements that are signed off by the customer.
- Having formal milestones encourages a disciplined approach.
  - Well-defined formal milestones in waterfall allows progress to be easily assessed both by the team and the customer
- It encourages the project to be managed in a well-structured manner.
- Encourages extensive documentation at every stage. After the project is complete, this documentation can be reviewed. Leaders of future projects can learn from previous projects - what worked and what didn’t.
- Suited for projects that are well understood in terms of technology risk and have clear and stable requirements.
- Easy for someone outside the project (for example an auditor) to understand how the project is managed and progressed.

Disadvantages

- Not so good for high-risk or challenging projects. It is difficult to plan ahead if the team is unfamiliar with the technology or techniques required.
- Requiring extensive documentation and tight discipline can stifle the creativity and individuality of the programmers.
- Not so suitable when the requirements are vague or constantly being changed
- Not such a good fit for companies that prefer a more informal way of running projects.

ITERATIVE

An iterative life cycle model does not attempt to start with a full specification of requirements. Instead, development begins by specifying and implementing just part of the software, which can then be reviewed in order to identify further requirements. This process is then repeated, producing a new version of the software for each cycle of the model.



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Advantages

- We can create a high-level design of the application before we actually begin to build the product and define the design solution for the entire product. Later on we can design and built a skeleton version of that, and then evolve the design based on what has been built.
- We are constantly building and improving the product step by step. Hence we can track any problems in the early stages. Thus avoiding the downward flow of the defects.
- We can get reliable user feedback as the system is developed. When presenting sketches and blueprints of the product to users for their feedback, we are effectively asking them to imagine how the product will work.
- In iterative model less time is spent on documenting and more time is given for designing.

Disadvantages

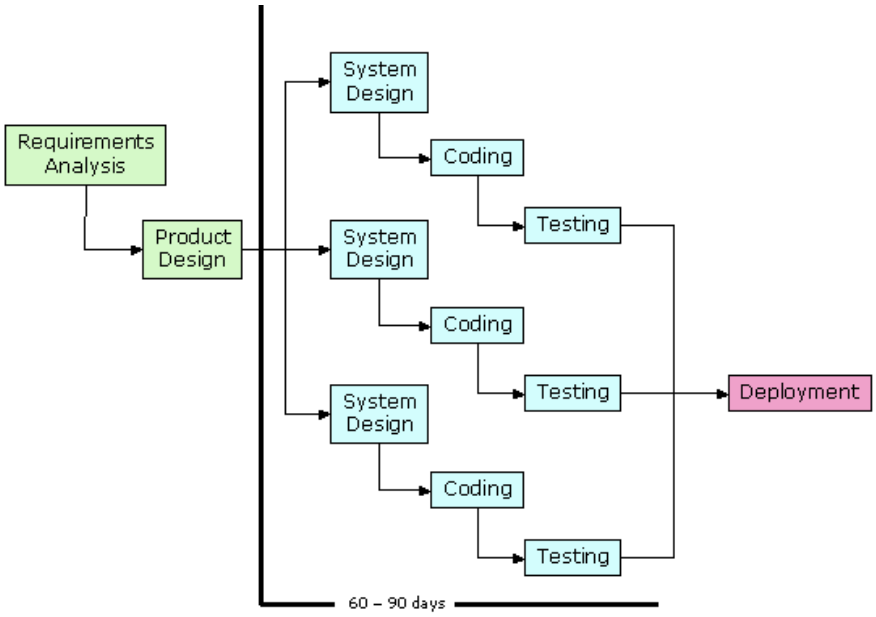
- Each phase of an iteration is rigid with no overlaps
- Costly system architecture or design issues may arise because not all requirements are gathered up front for the entire lifecycle

When to use it

- Requirements of the complete system are clearly defined and understood.
- When the project is big.
- Major requirements must be defined; however, some details can evolve with time.

RAPID APPLICATION DEVELOPMENT (RAD)

RAD or Rapid Application Development model is a type of incremental model. In RAD model the components or functions are developed in parallel as if they were mini projects. The developments are time boxed, delivered and then assembled into a working prototype. This can quickly give the customer something to see and use and to provide feedback regarding the delivery and their requirements.



Advantages

- Reduced development time
- Increases reusability of components
- Quick initial reviews occur
- Encourages customer feedback
- Integration from the very beginning solves a lot of integration issues.

Disadvantages

- Depends on strong team and individual performances for identifying business requirements.
- Only systems that can be modularised can be built using RAD
- Requires highly skilled developers/designers.
- High dependency on modelling skills
- Inapplicable to cheaper projects as cost of modelling and automated code generation is very high.

When to use it

It should be used if there's high availability of designers for modelling and the budget is high enough to afford their cost along with the cost of automated code generating tools RAD SDLC model should be chosen only if resources with high business knowledge are available and there is a need to produce the system in a short span of time (2-3 months).

STAGES OF THE SDLC

PRELIMINARY ANALYSIS

- Find the organisation's objective as well as the scope & nature of problem (Problem Definition)
- See how the problem fits in with the organisation's objectives
- Propose alternative solutions: leave system as is, improve it, or make a shiny new one

Feasibility Study

1. **Technical feasibility:** Does the company have the technical resources/knowledge needed for the project?
2. **Economic feasibility:** Checking if the company can afford the project and if the benefits outweigh or are worth the cost
3. **Legal feasibility:** Making sure the company is able to complete the project through legal means
4. **Operational feasibility:** Measure of how well a company can solve project problems and take advantage of opportunities over the course of the project
5. **Scheduling feasibility:** Checking if the company has enough time to do the project and that it won't take too long to finish

ANALYSIS

- Define project goals into clear functions + operation of intended software
- Consider functional + non-functional requirements
- Look @ how requirements will b fulfilled
- Look @ how current system functions + good and bad parts of it

DESIGN

- Describes wanted features and operations in detail
- Screen layouts, business rules, process diagrams, pseudo code + other documentation
- Technical details of design discussed with the customer
- Discuss business needs
- **Physical design**: the technology-specific details from which all programming and system construction can be accomplished
- **Logical design:** Refers to theoretical representation of data flows, inputs and outputs of system.
  - Usually done by modelling using visual model of system.
  - Includes ERDs (Entity Relationship Diagrams)

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DEVELOPMENT

- Hardware and software acquisition
- Construction and testing: start building new system and testing it to make sure it works properly
- Coding done here
- Unit testing of process by dev team
- After each stage developers can improve and tweak their work
- Developers must be open minded and flexible
- Concept phase

IMPLEMENTATION

- **Software put into production** (may need 2 train staff)
- Roll out either in stages or full blown

Change-over methods

- **Direct cut:** Once new system ready, whole org transfers on 1 day to it
- **Phased:** Phasing into new system one step @ a time
- **Pilot:** One section/ department of org is transferred to new system. Any bugs are fixed b4 whole organisation release (Think pilot episode or beta testing)
- **Parallel:** Old + new system run side-by-side for short amount of time. Both r maintained + kept up 2 date so if something happens 2 new system u can go back 2 old one.
  - Is expensive and very labour intensive

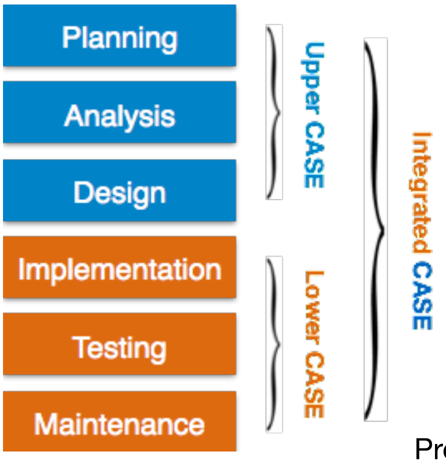
EVALUATION AND MAINTENANCE

- **Termed support & monitoring performance**
- **Evaluate efficiency of system and standards**
- **Performance evaluation:** performance of new system is assessed + recorded (to help with improvement of next system you'll work on)
- **Fault finding and correction:** Finding issues with system and fixing them throughout rest of system life

DATA GATHERING TECHNIQUES

- **Observation:** When we simply look at the existing system and take notes on how it functions
- **Questionnaire:** We can create surveys or questionnaires to give to users of the system to get some realtime feedback and information regarding the system
- **Interview:** We can also interview existing users to examine their knowledge of the current system, and also to gather feedback on potential changes or upgrades
- **Sample forms:** We can create forms to get information from current users, similar to surveys and questionnaires
- **Sampling volume of work processed by system:** Also known as studying documentation, we can simply take existing documents and work and write notes based off the information provided

PROJECT MANAGEMENT CASE TOOLS



Central Repository: Serves as central storage point that holds product specifications, requirement documents, related reports, and data diagrams. Also acts as data dictionary.

DIAGRAM TOOLS

Used to represent system components, data and control flow, and system structure in graphical form.
EX: Flow Chart Maker

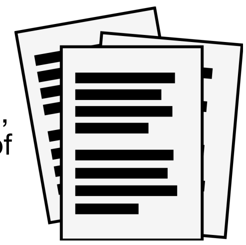
PROCESS MODELLING TOOLS

Process modelling is method to create software process model, which is used to develop the software. Process modelling tools help managers choose process model or modify it as per the requirement of software product.
EX: EPF Composer

PROJECT MANAGEMENT TOOLS

Used for project planning, cost and effort estimation, project scheduling, and resource planning. Help in storing and sharing project information in real-time throughout the organisation.
EX: Creative Pro Office, Trac Project

DOCUMENTATION TOOLS



Generates documents for technical and end users.
Technical users refer to the system manual, reference manual, training manual, and installation manuals.End user documents focus on functioning & how to of system such as user manual.
EX: Doxygen, & Dr Explain

ANALYSIS TOOLS

Tools that help gather requirements, & auto check for errors or faults.
EX: Accept 360, Accompa

DESIGN TOOLS

Help software designers design block structure of software which can be broken down further into smaller chunks. Provide detail of each module & interconnections amongst modules.
EX: Animated Software Design

CONFIGURATION MANAGEMENT TOOLS

Deals with version & revision management, baseline configuration management, and change control management. Helps by auto tracking as well as version and release management.
EX: Fossil, Accu REV

CHANGE CONTROL TOOLS

Considered part of configuration management tools.
Deals with changes made to the software after baseline is fixed or when it's 1st released. Tracks changes & also helps enforce the change policy of the organisation.

PROGRAMMING TOOLS

These tools consist of programming environments like IDE (Integrated Development Environment), in-built modules library and simulation tools. These tools provide comprehensive aid in building software product and include features for simulation and testing.
EX: Cscope to search code in C, Eclipse

PROTOTYPING TOOLS

Prototype provides initial look and feel of the product and simulates few aspect of actual product.
Prototyping CASE tools essentially come with graphical libraries. They can create hardware independent user interfaces and design. These tools help us to build rapid prototypes based on existing information. In addition, they provide simulation of software prototype.
EX: Serena prototype composer, Mockup Builder

WEB DEVELOPMENT TOOLS

These tools assist in designing web pages with all allied elements like forms, text, script, graphic and so on. Web tools also provide live preview of what is being developed and how will it look after completion.
EX: Fontello, Adobe Edge Inspect, Foundation 3, Brackets

QUALITY ASSURANCE TOOLS

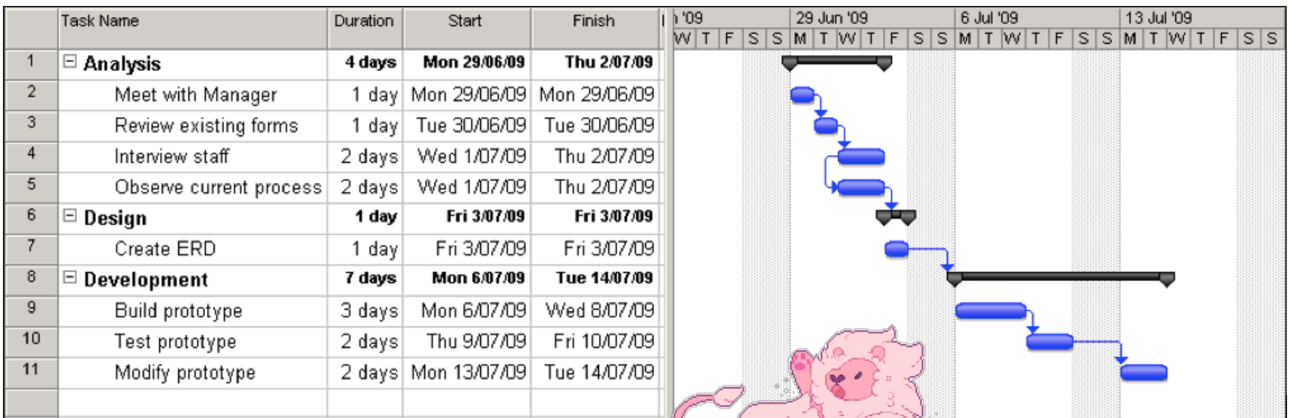
Monitoring engineering process & methods adopted to make software product in order to make sure quality matches the organisation's standard. Consists of configuration, change control, and software testing tools
EX: Soap Test & AppsWatch

MAINTENANCE TOOLS

Includes software changes after delivery. Auto logging and error reporting techniques, auto error ticket generation, & root cause analysis
EX: Bugzilla, and HP Quality Centre

GANTT CHART

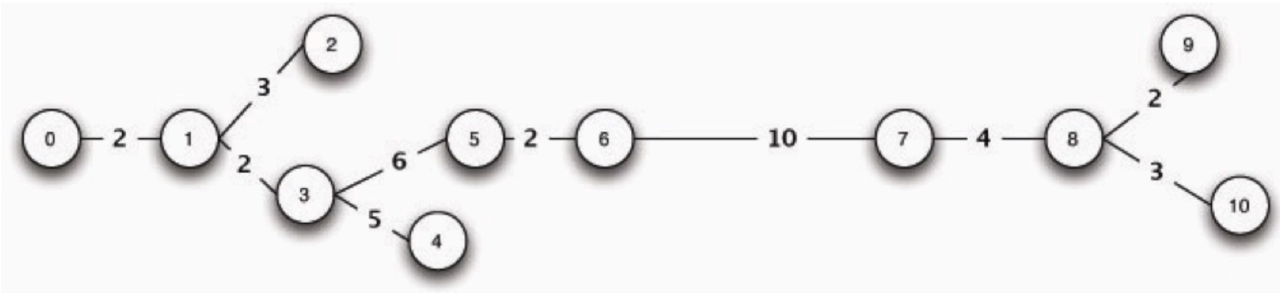
Shows what needs to be done and when it's supposed to be done. A way of showing tasks and events displayed against time. Each task represented by a bar. Position and length of the bar shows the start date, time length, and end date. Use when scheduling and monitoring tasks within project. Also use when steps, their length and sequence are known.



	Task Name	Duration	Start	Finish
1	**Analysis**	**4 days**	**Mon 29/06/09**	**Thu 2/07/09**
2	Meet with Manager	1 day	Mon 29/06/09	Mon 29/06/09
3	Review existing forms	1 day	Tue 30/06/09	Tue 30/06/09
4	Interview staff	2 days	Wed 1/07/09	Thu 2/07/09
5	Observe current process	2 days	Wed 1/07/09	Thu 2/07/09
6	**Design**	**1 day**	**Fri 3/07/09**	**Fri 3/07/09**
7	Create ERD	1 day	Fri 3/07/09	Fri 3/07/09
8	**Development**	**7 days**	**Mon 6/07/09**	**Tue 14/07/09**
9	Build prototype	3 days	Mon 6/07/09	Wed 8/07/09
10	Test prototype	2 days	Thu 9/07/09	Fri 10/07/09
11	Modify prototype	2 days	Mon 13/07/09	Tue 14/07/09

PERT (PROGRAM EVALUATION REVIEW TECHNIQUE) CHART

Program management tool used to schedule, organise, and coordinate task within project. Use when managing a project and when you need to clearly show task dependencies.



SYSTEMS DEVELOPMENT DOCUMENTATION AS PART OF THE SDLC

SYSTEM MANUALS

System manuals can also be referred to as internal documentation. Internal documentation refers to notes and instructions within projects, often seen only by developers. This includes comments within the source codes of a project. System manuals are generally created for developers and people within the project

USER MANUALS

User manuals can also be referred to as external documentation. External documentation is any form of manual or guide used to instruct the end-user of a system. User manuals are generally created for end-users in order to educate them on how to use the system.

STANDARD OPERATING ENVIRONMENT

PURPOSE

To have a standardised and managed computing environment.

ADVANTAGES

-It's easy to upgrade and maintain software and hardware on multiple computers as they're identical.
-Costs are significantly cut as bulk software licensing and hardware purchases can be made.
-Time to install and maintain the systems is shortened through processes such as Imaging
-Essential upgrades and fixes can be tested on a small number of systems before rolling out to the network, hence cutting downtime.

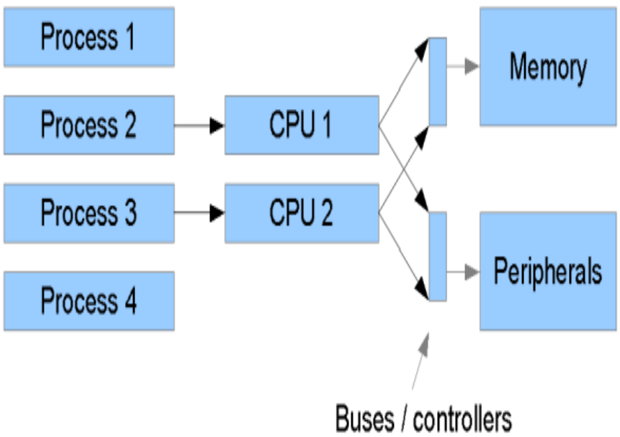
ROLES OF OPERATING SYSTEMS

SCHEDULING

- **Scheduling:** process of optimisation, control and arrangement processes and data handled by the CPU
- There are a set of vital time tasks which need to be considered in the process of scheduling:
- **Arrival Time:** Time at which the process arrives in the ready queue
- **Completion Time:** Time at which process completes its execution
- **Burst Time:** Time required by a process for CPU execution
- **Turn Around Time:** Time difference between completion time and arrival time
- **Waiting Time:** Time difference between **turn around time** and **Burst Time**

MANAGING CONCURRENCY

- One of the main task handled by the CPU scheduler is concurrency management.
- **EX:** task x and task y both need to be processed independently so task x starts running and y starts running before x has finished.
- Concurrency can be handled in a variety of ways. Most common method is through multiple core CPU's connected at the bus level
- Another way to manage concurrency is by task switching.



MANAGING MEMORY

- Memory management refers to management of Primary Memory/Main Memory
- Main memory accessed directly by the CPU.
- The OS:
  - Keeps tracks of primary memory
  - In multiprogramming, the OS decides which process will get memory, when and how much.
  - Allocates requested memory.
  - De-allocates the memory when a process has been terminated or completed.

MANAGING DEVICES

- An Operating System manages device communication via their respective drivers. It is responsible for managing the processes for the following devices:
- Keeps tracks of all devices. Program responsible for this task is known as the **I/O controller**.
- Decides which process gets the device when and for how much time.
- Allocates the device in the efficient way.
- De-allocates devices.

FILE SYSTEMS

ROLE OF FILE SYSTEMS

The role of file systems is to manage and store user data.

FEATURES OF FILE SYSTEMS

Space Management

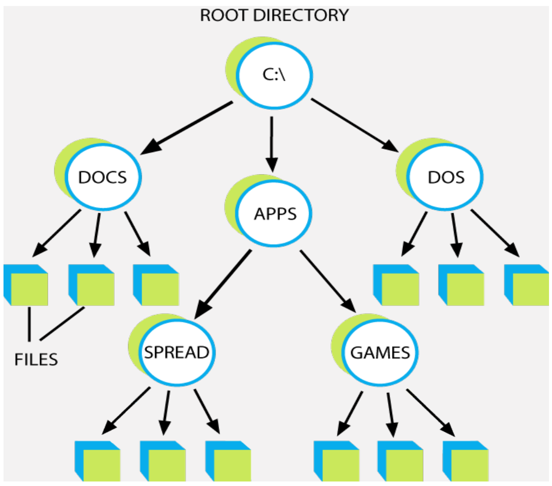
- Space management refers to how we use and manage space on our storage devices
- Often, storage devices such as hard disks (HDDs) and solid-state drives (SSDs) are **partitioned**
- **Partitioning:** action of breaking up a storage device in to many usable chunks
  - This has many benefits, including:
    - Option to run multiple OS (each OS installed on a different partition)
    - Partitions can be used for storage space outside of an OS, meaning even if the OS corrupts, the data on the created storage partition will remain
    - Partitioning eliminates need for multiple physical storage devices as we can divide one storage device in to many *virtual* storage devices
- Operating Systems will often create multiple partitions when installing, however we don't generally see these partitions on common Operating Systems such as Mac OS X and Microsoft Windows
  - These system-created partitions are generally for use as reserved sections that allow for important data
  - This generally includes a boot manager and necessary configuration files for booting in to the OS
- Operating Systems contain disk utility applications that allow for creating, deleting and modifying partitions

File Names

- **File names:** used to identity where a specific set of information is stored on a file system
- A file name is essentially an identifying name given to a computer file
- Some systems are case sensitive, meaning a file named "MyFile" and "myfile" point to different files
- Generally systems are case insensitive, meaning "MyFile" and "myfile" will point to the same file
- Special characters such as ':' aren't generally allowed in file names

Directories

- File systems contain directories, or more commonly referred to as folders
- These directories are capable of grouping specific files in to a centralised place
- Work off a hierarchy basis, meaning each folder can contain multiple sub-folders, each containing their own files



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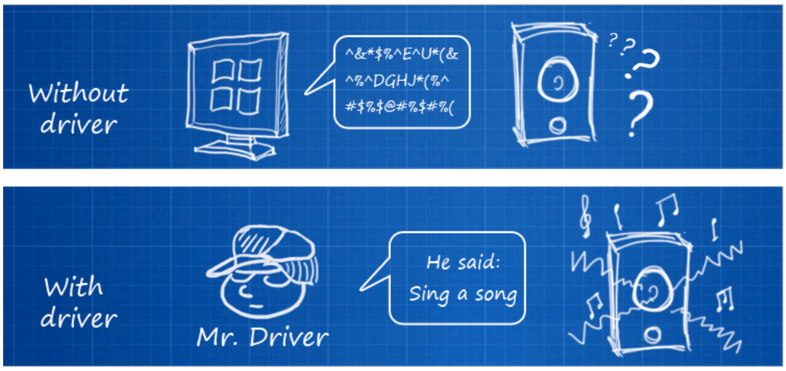
Role of Drivers

- A device driver/driver: program that controls devices attached to a computer system. Drivers act as a translator between a piece of hardware and the operating system.
- Operating systems often come with many pre-installed drivers in order to function. Common drivers are for graphics card, network interface cards and even drivers to support peripherals such as your mouse and keyboard.
- Essentially, a driver is used to convert general input and output commands from the operating systems to instructions that the device, such as a printer or display, can understand.



TYPES OF OPERATING SYSTEMS

EMBEDDED

- Specialised OS used in computers built into larger systems
- Designed to be compact, resource usage efficient, & reliable
- DOESN'T load + execute apps
- Therefore, system only able to run single application
- Usually used for hardware that have very little computing power
- Apps r usually built into/part of OS so they load immediately when OS starts
- **EX:** iOS, windows mobile, android

STAND ALONE

- An OS that works on a computer that is by itself
- Doesn't need any library or other OS to boot up
- **EX:** Windows 7 and Mac OS X
- Basic criteria for a standalone system includes:
- An OS that never exits
- An OS that loads in to memory
- An OS that begins its own execution
- An OS that never fully hands over execution

SERVER

- Is designed specifically for a task (usually over a network)
- Has special features and abilities required by a client-server architecture
- **EX:** Mac OSX server and Windows small business server

COMPONENTS OF THE CENTRAL PROCESSING UNIT

ARITHMETIC LOGIC UNIT (ALU)

- Complex digital circuit
- Carries out arithmetic & logic operations on operands (data that can b operated on)
- Some CPUs have ALU separated into AU and LU
- A.K.A integer unit (IU)
- Logic operations are things like is 1 > 2=FALSE

CONTROL UNIT (CU)

- Controls all other units in computer system
- Controls flow of data between CPU and other devices
- Let's ALU, memory, input and output devices know how to respond to instructions from a program
- Completes fetch-execute cycle
- Interprets instructions by changing them into format that processor needs to perform the function
- Controls timing signals for computers
- This means it's responsible for pulse of info packet, to make sure they are sent out regularly (every few milliseconds)

REGISTERS

- Temporary storage of data
- Can hold all data types
- Must be large enough to store an instruction EX: In a 64-bit computer the register must be 64-bits long
- All data must be stored in a register before it is processed

PROGRAM COUNTER

-Type of register that contains address of instruction currently being executed
-Increases by 1 after fetching an execution
-A.K.A. instruction address register or address pointer

SYSTEM CLOCK

-Used to set tasks and processes (synchronisation + scheduling)
-Vibrates at a specific frequency
-Sped of computer processor measured in clock speed
-Shortest time any computer is capable of performing is 1 clock or 1 vibration of clock chip
-Because the entire system is linked to the clock, speeding it up has greater improvement on system speed than increasing raw process speed

DATA,ADDRESS AND CONTROL BUS

- Buses are a set of physical connections that can be shared by many different pieces of hardware so they can connect to each other
- They exist to reduce number of paths for hardware communication by carrying them all on one path
- Data buses transfer instructions to and from the processor (moves in two directions)
- Address buses move memory addresses needed by the processor to read and write data (only moves in one direction)
- Control buses transport orders and synchronisation signals coming from the control unit and moving to other hardware components (moves in two directions)

FETCH EXECUTE CYCLE

PURPOSE

To obtain and complete instructions that the computer needs to complete a task.

STAGES OF FETCH EXECUTE CYCLE

Fetch

- Instruction fetched from memory address currently stored in program counter.
- At the end of fetch operation, the program counter points to the next instruction that will be read next cycle.

Decode

- The decoder in the control unit works out (decodes) what to do with the instruction.
- Each CPU is designed to understand a specific set of commands.
- These commands are referred to as the instruction set.
- These instruction sets work by providing the CPU with machine code (hexadecimal or binary instructions)

Execute

- When the instruction has been decoded the CPU can execute the instruction
- The control unit passes instruction as control signal sequence to the proper function units to carry out instructions
- If the ALU is involved, it sends conditional signal back to control unit
- Operation result is stored in main memory sent to output device

Store

- Send and write the results back in main memory.
- The newly processed data is then passed back in to memory fresh from its processing via the data bus. The data can be re-submitted for further processing or be used for a practical use, such as reporting and other outputs.

PROCESSING

PURPOSE OF PROCESSOR ARCHITECTURES FOR DIFFERENT TYPES OF SYSTEMS

- A **processor architecture** refers to the complex art of designing and constructing CPUs
  - Its purpose is compactness and efficiency
- RISC (reduced instruction set computers) is one type of CPU structure.
  - used in high end CPUs for applications that need video processing, supercomputers etc
  - has no memory unit and uses separate hardware to process instructions
  - Advanced RISC Machine (ARM) is most common in many smartphones
  - very fast
- CISC (complex instruction set computers) is another type of CPU structure
  - used in low end CPUs for applications like modern cars, home security, home automation such as vacuum cleaners, smart fridges
  - has a memory unit to conduct complex instructions
  - not so fast

TYPES OF PROCESSING

Distributed processing

Distributed processing is a setup in which multiple individual central processing units (CPU) work on the same programs, functions or systems to provide more capability for a computer or other device.

Sequential Processing

Sequential processing is a term used to describe the processing that occurs in the order that it is received.

Parallel Processing

Parallel processing is the method of evenly distributing computer processes between two or more computer processors. This requires a computer with two or more processors installed and enabled. It also requires an operating system capable of supporting two or more processors, and software programs capable of evenly distributing processes between them.

Purpose of Using Benchmark to Determine System Performance:

To compare and test the performance of different systems. This is because it's hard to compare system performance based on specifications alone.

Multi-core Processing

-Single processor incorporates more than one processor core
-Completes multiple tasks at the same time
-Each of these cores acts as an individual processing device
-Often, modern computers run on dual-core and quad-core processors
-Multi-core processing allows for tasks to be run simultaneously, meaning that more than one task can be processed at a time

DISASTER RECOVERY

DISASTER RECOVERY PLAN

- A DRP is a preplanned set of actions that are to be taken in the event that a disaster affects the technological infrastructure of an organisation.
- The purpose of a DRP is to ensure that the computer systems are stable and able to operate fully after a disaster

ONLINE STORAGE

- Copies of data are sent over a network to a server that is located off site
- The server is hosted and maintained by a dedicated data storage company/service provider
- The service is payed for and the costs are based on the capacity, bandwidth, or number of users

Advantages	Disadvantages
- Cost efficient - Almost unlimited storage - Reliability - Manageability - Accesability	- Security - Speed - Requires network connection

INCREMENTAL BACKUP

- Only backs up changes since the previous incremental or full backup
- Allows backups to be completed faster and with less storage

Advantages	Disadvantages
- Fast backup windows - Less storage space needed - Allows retention of several versions of the same file	- Slower recovery - Initial full backup is required before incremental backups can start - Takes longer to restore a specific file - If one backup fails the recovery will be incomplete

FULL BACKUP

- It's the starting point of all other backups
- You must do a full backup before doing any other types of backups
- With this the entire system including updates and previous data versions are backed up

Advantages	Disadvantages
- Fast and easy data recovery because data is readily available - Files and folders are backed up to one backup set - Easy version control	- Needs more storage space - Additional bandwidth is required - Time consuming if full backup is run all the time

RAID (Level 0,1,10)

- **RAID:** Redundant array of inexpensive/independent disks
- Organised into level 0 to 9 but two levels can be combined to create 2 digit levels
- Uses techniques like disk striping and disk mirroring
  - **Disk striping:** dividing data into blocks and spreading the blocks across multiple storage devices (RAID level 0)
  - **Disk mirroring:** replication of two or more disks (RAID level 1)
- RAID level 10 is a combo of level 0 and level 1
  - It strips data across mirrored pairs
- RAID consistently distributes data across each drive in the array then breaks the data into consistently sized chunks
  - Each chuck is then written to a hard drive in the RAID array based on the level used
  - When the data is read, the process is reversed, giving the illusion that the multiple drives in the array are actually one large drive

RAID 0

- Needs at least 2 disks to work
- Suitable for audio and video streaming and editing, high end gaming or hobbyist systems, data that changes infrequently and is backed up regularly needing high speed

Advantages	Disadvantages
- Increased performance (write and read data speeds) - Cheapest RAID level - Supported by lots of hardware and software	- Drive failure results in complete data loss short of specialised data recovery - If one disk fails that affects the whole array and the chance of data loss or corruption increases

RAID 1

- Needs at least 2 disks to work properly
- Suitable for any application that needs a very high availability, mission critical storage, small servers that only use 2 data drives

Advantages	Disadvantages
- Fault tolerance - Easy data recovery - Data copied seamlessly and simultaneously - If a disk is fried the other(s) can still work - Relative low cost	- Causes slight drag in performance - Lower user capacity

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RAID 10

- Requires at least 4 disks to set up
- Suitable for mission critical tasks like accounting systems

Advantages	Disadvantages
- Gives best performance - If something goes wrong in one disk the rebuild time is very fast since all that's needed is copying data from surviving mirror to the new drive - Secure because mirroring duplicates data - Fast - Chunks of data can be read and written to different disks simultaneously	- Expensive - Complex to set up

UNINTERRUPTIBLE POWER SUPPLY (UPS)

- A device that provides instant power cutover when utility power fails
- Provides a backup electrical power supply ensuring continuous power supply to important equipment if the original power supply is lost
- Often supplied as a bank of batteries that offer short term protection against power surges and outages
- Usually only allows enough time to turn critical systems off

VIRTUALISATION

BENEFITS OF VIRTUALISATION

-Virtualisation allows for singular hardware resources to run multiple software resources.
-This cuts down initial costs as additional hardware purchases aren't necessary.
-Hardware is better utilised as often one application or process won't use the full potential of the hardware.
-Managing users and applications on a virtual server can be far easier than managing multiple servers with singular users and applications.
-Faster application and service delivery.
-Energy savings as less physical machines means less power and conditioning needed.

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TYPES OF PLATFORM VIRTUALISATION

Desktop/PC Virtualisation

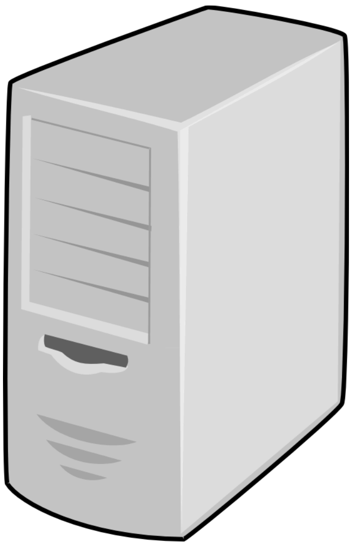
- Virtualisation tech used to isolate computer desktops from physical computers
- EX: VMware to run windows on a Mac
- Separates desktop environment from physical device and configured as virtual desktop infrastructure (virtualisation technology that hosts a desktop operating system on a centralised server in a data centre)

Advantages	Disadvantages
- Users able to access all personal files and applications from any location and on any PC - Lowers cost of licensing for installing software on desktops and maintenance is simple - Provides a way for users 2 maintain their individual desktops on single central server - Users may be connected to central server via LAN, WAN, or over internet	- Difficulty of maintenance + set up of printer drivers - Increased downtime in case of network failures - Complexity + costs involved in VDI deployment

Server Virtualisation

- Involves division of physical server into small virtual servers using virtualisation software
- Ever server runs multiple Operating Systems at the same time

Advantages	Disadvantages
- Reduced costs - Many operations can be automated - Has features that help with backup and recovery - Centralised management - Can use OS that isn't compatible with hardware	- Has high upfront costs - Security risks because of how host company protects data - Takes a while to learn how to use - Needs monitoring and configuration - Uses a lot of disk space - Uses up a lot of RAM



There are multiple types of server virtualisation:

- **Para virtualisation:** also uses virtual machines. However virtual machine monitor (VMM) changes code of guest's OS (porting)
- **OS virtualisation:** The host runs a single OS kernel as core & exports OS functionality to each guest. Guest MUST use same OS as host
- **Complete virtualisation:** Uses a virtual machine monitor that interacts with the physical server CPU and disk space. Allows multiple OS's run at the same time on the host computer whilst each virtual server is independent of the other virtual servers running

Storage Virtualisation

- Very easy and cost effective to implement
- Handy for disaster recovery since data stored on virtual storage can b replicated and transferred to another location
- Process of grouping physical storage from multiple network storage devices so that it looks like single storage device
- By consolidating storage into centralised system u can eliminate costs of managing multi storage devices

Advantages	Disadvantages
- Less energy use - More disk space - Cost-effective because you don't have to buy a lot of additional software - Increased loading and backup speed - Same amount of work can be done with less servers needed	- Often violates licensing agreements - Reduced costs + increase in disk space can encourage ppl 2 increase server number making too many servers to manage - Network system is much more complicated - If one system fails they all fail - If one server is infected/breached, entire network is compromised

Application Virtualisation

- Separation of an installation of an application from the computer accessing it
- Fools the computer into working like the application is being run locally on the computer accessing it
- There are 2 types: remote and streaming
  - Remote applications run on a server and end users view and interact with the apps over a network using a remote display protocol
  - With streaming applications the virtualisation app runs on the end user's computer and, when the app is being used, parts are downloaded to the end user's computer on demand

CLOUD COMPUTING

Cloud computing refers to accessing data and programs over the Internet as opposed to your Computer's hard-drive
It can be seen as relying on your Internet connection to store, retrieve and even process your data

Purpose

Allows users to run particular software, as well as have access to their stored data anytime, anywhere and on any computing device.

Advantages

- Cost efficiency
  - Users can save money on storage devices as their data is hosted externally
  - Program licensing costs can be reduced as software can be hosted and used externally
- Convenience
  - The cloud can be accessed wherever there is an Internet connection available
  - Users have peace of mind knowing their data is hosted externally, as it eliminates the risk of leaving data at home or in another environment
  - Data is available whenever and wherever its needed
- Backups & Recovery
  - Backups can be made online without hassle
  - Recovery is easy as data is stored online and can be accessed from anywhere
- Storage Capacity
  - The Cloud often offers more data storage capacity compared to physical hardware
  - Cloud servers have a large amount of storage available
  - The cost for storage is much, much cheaper than the cost for an equal-sized physical storage device

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Disadvantages

- Privacy & Security
  - As your data is hosted elsewhere, it's not easy to ensure the privacy and protection of it
  - Each company has different policies on how they store your data and what it can be used for
  - Security is another factor that needs to be addressed, as personal information because available to the world wide web
- Downtimes and Unavailability
  - Sometimes hosts can go down
  - This means that your data may be inaccessible if a company hosting your storage has a downtime
- Limited Functionality
  - In terms of programs and hardware, functionality is limited to the company's discretion
  - Features and performance of the hardware is dependent on the company's architecture
  - As many users will often use the same resources, certain aspects of the cloud may be slow

CONVERGENCE OF TECHNOLOGIES, INCLUDING THE CONTINUED DEVELOPMENT OF ICT SYSTEMS

WHAT IS IT?

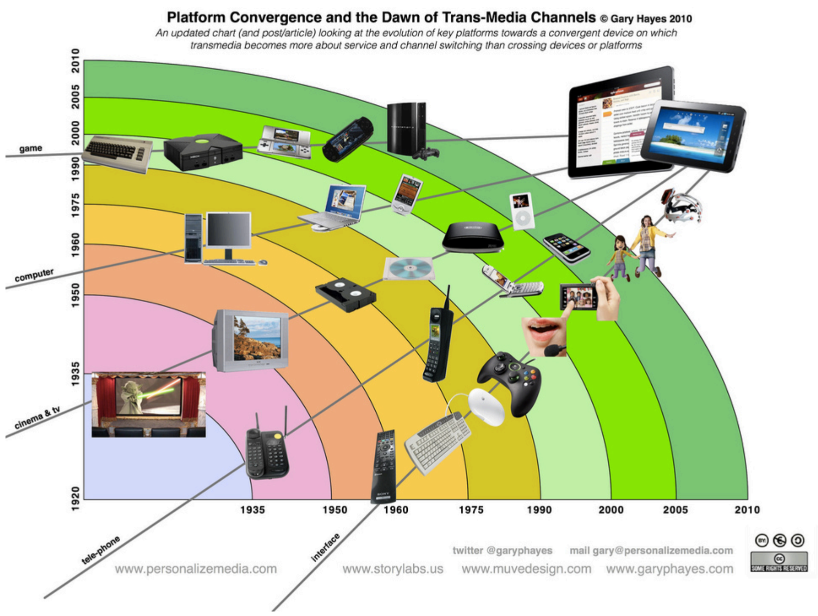
- In technology, products **converge** when they evolve to perform similar tasks
- EX: A smartphone capable of taking photos (camera + phone = camera phone)

WHY IS IT IMPORTANT?

- As technology evolves, more and more features and tasks need to be addressed
- Instead of creating individual devices for each of these tasks, we can create one device to perform multiple
- Convenience is priority, a smartphone is capable of handling the tasks of a camera, telephone and a computer (to an extent)

MOBILE DEVICES & CONVERGENCE

- As mentioned, convenience is a large factor in **convergence**
- The ability to carry a single pocket-sized device with the capability of 4 devices is highly sought after in today's world
- Mobile devices such as smartphones and tablets are very good at handling these types of tasks that are common in other devices
- Such as: taking photos, browsing the Internet, playing games, taking notes, taking calls & messaging, viewing emails, etc.
- Smart Watches and TVs are the latest in technological convergence, the former allowing for these functions from your own wearable device



DISPOSAL OF DATA AND COMPUTER COMPONENTS

METHODS FOR THE SECURE DISPOSAL OF DATA

Physical Destruction of Media

- Shredding using company issued shredders.
- Placed in locked shredding bins for private contractor to come on-site and shred, witnessed by the company personnel throughout the entire process.
- Incineration using organisation incinerators or witnessed by company personnel onsite at agency or at contractor incineration site, if conducted by non-authorised personnel.

ENVIRONMENTAL ISSUES, RELATED TO THE DISPOSAL OF DATA COMPONENTS

Many electronics contain harmful chemicals that can negatively affect the environment. Most computer components are recyclable so you shouldn't throw away your old electronics in a regular bin or regular recycling bin.

Overwriting

Overwriting (at least 3 times) - an effective method of clearing data from magnetic media. As the name implies, overwriting uses a program to write (1s, 0s, or a combination of both) onto the location of the media where the file to be sanitised is located.

SYSTEMS ANALYSIS AND DEVELOPMENT LEGAL

PURPOSE OF INTELLECTUAL PROPERTY IN THE DEVELOPMENT OF ICT SYSTEMS

Intellectual Property

- A broad term that is used to describe the results of creative and innovative endeavours. It is described as 'intellectual' because it is the result of the application of the mind.
- It is described as 'property' because, just like other property, it can be owned, sold and transferred, leased or given away.
- Protected by law through patents, copyrights, and trademarks

Purpose in ICT

- IP is essential in protecting original ideas and creations
- New ideas arise daily in the ICT business
- IP allows us to protect these ideas by law and gain compensation if our creations are stolen

ROLE OF LAW AND ETHICS IN THE USE OF ICT SYSTEMS

Use of Code of Conduct Policies

- **Code of conduct:** set of rules used to set a standard of behaviour and social norms
- Each organisation will have a different code of conduct that outlines their particular practices, responsibilities and expected behaviour from its members and / or users
- A code of conduct is put in place to ensure harmony within an organisation as it encourages proper etiquette and behaviour when dealing with one another
- Common agreements outlined in a typical code of conduct relating to ICT are:
- All computers must be used for *[organisation]* purposes only
- *[Individuals]* must not disclose passwords to other individuals
- Non-educational games and software must not be installed on a computer

- Computer hardware should not be defaced or vandalised
- The network administrator has right to deleted anything deemed inappropriate on *[the organisation]*'s network (such as non-work / education related files)
- Codes of conduct apply to websites and online resources as well as computing systems
- Many websites create their own specific codes of conduct with their own particular guidelines
- These guidelines work to create a happy, safe and healthy browsing experience

Prevention of Software and Information Piracy

- A lot of times, prevention of piracy is a difficult task
- **Licensing** is a legal method of ensuring your work won't be stolen, although pirates often disregard these legal documents
- Special encryption and software signing methods can be put in place to ensure validation of software
- Often demo versions are released together with full versions of software to help restrict piracy
- Internet validation can be used to ensure a license has been purchased before starting the software / game

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Managing Data

TYPES OF PHYSICAL STORAGE OF DATABASES

ONLINE

Database that’s accessed by the internet instead of being stored on a single computer.
Hosted on websites in the cloud.

Advantages	Disadvantages
→ Can be accessed anytime, anywhere in the world → They can hold infinite data → Monthly subscription makes it affordable → Collaboration is easier → Doesn’t need storage space	→ Server downtime → Difficult to search the database → Security risks → Can be complex and hard to use → Needs user training which is expensive → Time consuming to make

Local

Not accessible by the internet. Stored locally inside the application, program or computer.
Can also be kept on an external device like a storage drive or CD.

Advantages	Disadvantages
→ Security → Doesn't need internet connection → Easier to search than online databases	→ Less accessible → If the device it's hosted on has issues you could loose the database

Types of Databases

Distributed

A database that's separated into parts and stored in different locations physically.
The processing is shared by different database nodes.

Advantages	Disadvantages
→ Data can be stored on different systems → Can be accessed over different devices → If a computer fails the database is still safe because it's not all stored on one device	→ Because data is accessed from remote systems the performance is reduced → Network traffic is increased → Different data formats are used in different systems → Database optimisation is difficult

Centralised

Databases that are kept in a single physical location.
The location it's stored in is a centralised computer. EX: Desktop or server CPU or mainframe computer.

Advantages	Disadvantages
→ Easier to change or update because everything is in one location → Speed of operation → Easy to manage	→ If the centre goes down everything goes down

Data Warehouses Data Marts and Data Mining

Structure of Data Warehouses and Data Marts

Data Warehouse

→ A data warehouse is a huge store of data that collected from a variety of sources inside a company that's constantly updated.
→ The data is stored there for a long of time.
→ Data warehousing is when the structure of a data warehouse is implemented.

Data Marts

→ A small part of a data warehouse that's for use only for a small team inside an organisation.
→ It's still a database.
→ They are read only.

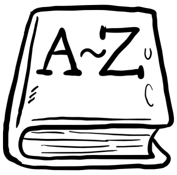
ROLE OF DATA MINING

- → Data mining is when you go through large amounts of data to turn it into information that you can use.
- → It's often used for analysing the habits of consumers
    - → This can be used to help with sales, marketing and promotions for products

ETHICAL IMPLICATIONS OF THE USE OF DATA WAREHOUSES, DATA MARTS AND DATA MINING

- → Privacy is the largest issue
- → Online companies hold large amounts of data on their users
- → These companies are able to sell your information to other companies for advertising in order for you to buy their products
- → If the systems companies use to hold your data are broken into malicious groups and entities can use your data for their own gain
    - → People could steal your identity and your money
    - → They could rack up lots of debt in your name and make large purchases like houses and holidays giving you a bad credit rating
    - → People could use obtained information to blackmail people also
- Loss of privacy
- Although used mainly for marketing products and services can allow ppl to access info about you that you don't want them to
- Even if you stop using online services like Facebook they can still hold onto and use your data

DATA DICTIONARY



PURPOSE OF A DATA DICTIONARY

- The main purpose of a data dictionary is to provide additional information about the data in a database (such as the data's attributes)
- Data dictionary's are especially important to developers, as they help maintain an up-to-date reference list items (and their properties) essential to the software
- In a relational database management system (RDBMS), a data dictionary also contains information on relationships between the given data and other data within a database

ELEMENTS OF A DATA DICTIONARY

- Element Name
- Data Type
- Size/Format Default
- Description
- Constraint

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DATABASE MANAGEMENT SYSTEMS CONCEPTS

DATA DEFINITION

- A data definition language (DDL) is a computer language used to create and modify the structure of database objects in a database. These database objects include views, schemas, tables, indexes, etc.
- It also describes the fields and records in a database table

Create

- CREATE: This command builds a new table and has a predefined syntax.
- The CREATE statement syntax is CREATE TABLE [table name] ([column definitions]) [table parameters].
- CREATE TABLE Employee (Employee Id INTEGER PRIMARY KEY, First name CHAR (50) NULL, Last name CHAR (75) NOT NULL).

Alter

- ALTER: An alter command modifies an existing database table.
- This command can add up additional column, drop existing columns and even change the data type of columns involved in a database table.
- An alter command syntax is ALTER object type object name parameters. ALTER TABLE Employee ADD DOB Date.

Drop

DROP: A drop command deletes a table, index or view. Drop statement syntax is DROP object type object name. DROP TABLE Employee.

DATA DUPLICATION

- Data duplication occurs when an exact copy of a piece of data is created
- For example, copy and pasting an item called "*MyPicture.jpg*"
  - The new pasted item contains the exact same data as the original picture

DATA INTEGRITY

Referential Integrity

- States that every foreign key must reference a valid existing value in another table
- This means that for every record in a normalised database, the linking element (the *foreign key*) must exist in another record
- Both the *primary and foreign keys* must be the same data type and length

Domain Integrity

- Ensures that data values inside database follow defined rules for value, range, and format

Entity Integrity

- A simple concept that ensures the validity of primary keys
- Ensures that each row in a table is a uniquely identifiable entity
- The concept states that each primary key must not be NULL (meaning it must contain a value of some sort)

DATA REDUNDANCY

- Data redundancy occurs when the same data is entered in to two or more fields of a database
- For example, "Joe" is entered in to the Name field under a record called *Customers*
- "Joe" is also entered in to the Customer field under a record called *Purchases*
- Although we are referring to the same Joe in both fields, each piece of data is seen as unique
- This means that to update "Joe", we need to manually edit each reference

DATA ANOMALIES

Insert Anomaly

When data can't be added to the database because other fields are missing data. Can't record information if whole row doesn't have data.

EmployeeID	SalesPerson	SalesOffice	OfficeNumber	Customer1	Customer2	Customer3
1003	Mary Smith	Chicago	312-555-1212	Ford	GM	
1004	John Hunt	New York	212-555-1212	Dell	HP	Apple
1005	Martin Hap	Chicago	312-555-1212	Boeing		
???	**???**	**Atlanta**	**312-555-1212**			

Delete Anomaly

Data is accidentally deleted when a row is deleted because the data is only stored in that row and no where else in the database.

EmployeeID	SalesPerson	SalesOffice	OfficeNumber	Customer1	Customer2	Customer3
1003	Mary Smith	Chicago	312-555-1212	Ford	GM	
~~1004~~	~~John Hunt~~	~~**New York**~~	~~**212-555-1212**~~	~~**Dell**~~	~~**HP**~~	~~**Apple**~~
1005	Martin Hap	Chicago	312-555-1212	Boeing		

Update Anomaly

When data is stored in multiple rows so when that piece of data needs to be updated it must be updated in each row. If this doesn't happen an error can occur.

EmployeeID	SalesPerson	SalesOffice	OfficeNumber	Customer1	Customer2	Customer3
1003	Mary Smith	Chicago	**312-555-1212**	Ford	GM	
1004	John Hunt	New York	212-555-1212	Dell	HP	Apple
1005	Martin Hap	Chicago	**312-555-1212**	Boeing		

DATA MANIPULATION

- Data Manipulation: the retrieval of information from the database
- insertion of new information into the database
- deletion of information in the database
- modification of information in the database

Common DML Commands

- **SELECT:** This command is used to output a list of rows (or record) from a database.
  - **SELECT** [x] FROM [y] WHERE [z]
- **UPDATE:** This command is used to alter the data from one or more tables.
  - **UPDATE** [x] SET [y] WHERE [z]
- **INSERT:** This command is used to add one or more entries to a database table.
  - **INSERT** INTO [x] [y] VALUES [z]
- **DELETE:** This command removes one or more entries from a database table depending on the conditions.
  - **DELETE** FROM [y] WHERE [z]

DATA SECURITY

- Refers to the collective measures used to protect and secure a database or database management software from illegitimate use and malicious threats and attacks.

Methods of Data Security

1. Physical security of the database server and backup equipment from theft and natural disasters
2. Reviewing existing system for any known or unknown vulnerabilities and defining and implementing a road map/plan to mitigate them
3. Restricting unauthorised access and use by implementing strong and multi-factor access and data management controls
4. Load/stress testing and capacity testing of a database to ensure it does not crash in a distributed denial of service (DDoS) attack or user overload

Common Data Security Vulnerabilities

1. Deployment failures
2. Broken databases
3. Data leaks
4. Stolen database backups
5. The abuse of database features
6. A lack of segregation
7. Database inconsistencies

NORMALISATION OF DATA TO THIRD NORMAL FORM

FIRST NORMAL FORM

The information is stored in a relational table and each column contains atomic values, and there are not repeating groups of columns.

SECOND NORMAL FORM

The table is in first normal form and all the columns depend on the table's primary key.

THIRD NORMAL FORM

The table is in second normal form and all of its columns are not transitively dependent on the primary key.

> It is simplest to think of transitive dependence as a column's value relying upon another column through a second intermediate column.

OPEN SYSTEMS

BACKGROUND

- In order to link to a website that provides a database it needs a system of connectivity.
- This connectivity is a special program called an Application Program Interface or API.

DATABASE CONNECTIVITY

- Picture 2 components here; you on the internet and a database on a computer.
- To get to that database it needs to made available on the internet.
- It can only be made available on the internet by using a special type of software called an API as mentioned above.
- It can also be made available by using Open DataBase Connectivity (ODBC) software.
- The API or ODBC make the connection of database to the internet happen.

DEVELOPMENT AND MANAGEMENT IN DATA DRIVEN WEBSITES

- A data driven website is the opposite of a static website that doesn't change. It's always changing.
- This is because a data driven website is used by eBusiness.
- Online businesses are examples of this.
- They want to update the items sold on the website.
- They want to update the prices on the website.
- Personal are employed to create the website, update all aspects of the site, and secure the website.
- There are web companies that can make data driven websites
- Management of these websites is software focussed.
- Software is used to analyse the data and return reports that provide guidance on clever business decisions.
- Online businesses and other successful large businesses are using reports to become even more successful.

Purpose of Database Documentation for the User:

The purpose of the database documentation is to help the person who is using the database.

ROLE OF LAW AND ETHICS IN STORAGE AND DISPOSAL OF PERSONAL DATA

- In Australia there are the Australian Privacy Principles contained in the 1988 Australian Privacy Act which instructs businesses and the government on how to store and use data.
- According to the Australian Privacy Principles:
    1. There should be open and transparent management of personal information from APP entities
    2. People should be given the option to not identify themselves or use a pseudonym when dealing with an APP entity
    3. APP entities shouldn't collect information unless its relevant or necessary for the entity's activities
    4. People must give consent for their information to be collected
    5. Information must be collected by legal methods, any information that isn't asked for or needed by the entity must be destroyed
    6. People must be notified if their information was collected by the entity from someone else or if they don't know that their information was collected
    7. People should be told why their information is being collected, who has access to their information, how they can access their information, the entity's privacy policy, and if the entity is likely to give their information to overseas entities

DESIGN CONSIDERATIONS

READABILITY

- Use words on the user interface that people can understand
- Technical language will turn people off.
- Short, sharp and to the point but effective.

NAVIGATION

- Navigation is logical and easy to follow
- Naming of menus is logical
- People expect to see particular navigation, show them normal navigation
- Provide tabs, menus in predictable places
- Provide focus for the most important item on the page/screen
- Provide alternative pathways, not just one pathway in and one pathway out
- Image navigation icons need to be able to work literally at the slide of a finger. Too small and it won't work
- Text hyperlinks too close together because of small font size can cause navigation problems

LOGICAL ORDER

- A clear and understandable interface is the goal
- Order items in the order in which people are going to use them
- Put first things up front and in the expected place
- Put yourself in the position of the person wanting to use this database, apply order to what they would expect

INCLUSIVITY

- Can all people use this software?
- Does it meet user friendly requirements regardless of :
  - culture
  - language
- Design it so it has maximum use by the people
- If people find it difficult to use, it won't be used very often.
- Design it with the end user in mind. All end users is the focus for inclusivity.

Congrats on finishing a topic! Keep going!!!

Developing Software

TYPES OF SOFTWARE LICENSE REQUIREMENTS

NETWORK (PER SEAT) LICENSE

Software license that's based on the number of users that have access to the software. Used in small companies because of the high cost. EX: A 100 user license based on users means that up to 100 specifically named users have access to the program

ENTERPRISE

An agreement to license the usage of a program to entire companies and organisations
It lets the organisation members have access to the software for a limited period of time. Used by large companies because of its size and affordability

Benefits

- There is a large cost saving because of volume discounts
- Consistent pricing over the licensing period
- Predetermined pricing discount for new purchases

END USER LICENSE AGREEMENT (EULA)

Licensing agreement between a software creator/publisher and the user. Similar to rental agreement (user pays to use the software copy NOT own it). User MUST accept EULA terms prior to using the software. EX: Use in video games and online purchases of software

COMMERCIAL/PROPRIETARY

Let's users use one or more copies of software but the copies are still owned by the publisher not the users.

FACTORS AFFECTING THE DEVELOPMENT OF SOFTWARE

USER NEEDS

The software must do what the user expects it to do. The users must be considered in the planning stages and they must be given opportunities to look at the design and give feedback on how well it will work for them.

USER INTERFACE

Includes input controls, navigation controls, and information components. The interface must be simple and user friendly so that users are able to fully utilise the software for what they need.

PROCESSING EFFICIENCY

Software has different memory and CPU intensity. When designing software, it must be designed to match particular hardware. These are listed in the technical specifications when purchasing the software.

DEVELOPMENT TIME

How much time is required to make the software and how many programmers are working on the project. If not enough time is given to developing the software then there is a chance that it won't work as intended or be filled with errors and bugs rendering it unusable.

TECHNICAL SPECIFICATIONS

Its important to match the software to the hardware (hardware software compatibility)
When advertising software, technical specifications must be listed.

DEVELOPING SOFTWARE LEGAL AND ETHICS

LEGAL OBLIGATIONS OF DEVELOPERS WHEN CREATING NEW SOFTWARE

- If you're making software for a company, the software is owned by the company, not you.
- You are required to protect and take care of data from users.
- When collecting user data, the data must be necessary for the company and must be collected legally.
- Users must be told why their data is being collected, the laws permitting the collection, and who will be given access to their data.
- Collected data must be kept up to date, relevant, and complete
- Collected data must be kept securely and safely
-

PROFESSIONAL ETHICS OF DEVELOPERS WHEN CREATING NEW SOFTWARE

- Contribute to society and human well-being
- Avoid harm to others
- Give proper credit for intellectual property
- Respect other people's privacy
- Approve software only if you believe it's safe and meets specifications
- Make sure that their software is tested, debugged, and reviewed properly, and to a high standard.

LEGAL AND ETHICAL RESPONSIBILITIES OF SOFTWARE USERS

- To pay for software that isn't freeware (commercial software)
- To not copy software if that isn't permitted under EULA
- To not gain pirated software and install it on their device
- To follow terms present in the software's EULA

STAGES OF THE SOFTWARE DEVELOPMENT CYCLE

ANALYSE DETAILED REQUIREMENTS

- What is required of this software?
- Who is going to use it?
- Why are they going to use it?
- When will they use it?
- What problem will the software overcome?

DESIGN DATA AND ALGORITHMS

- Plan a solution to the questions above.
- Create flowcharts and show the client to see if these will provide a workable solution.

CODE DATA STRUCTURES AND INSTRUCTIONS

- Do the programming.
- Do commenting within the code so other developers can understand exactly what the code does. Do this in small chunks at a time.
- Make the instructions for how the program will work. (use your commenting for this)

DEBUG SYNTAX AND LOGIC ERRORS

Run your programming to find any errors.

TEST TO MEET SPECIFICATIONS

- Select other test staff or people to run the program you made on exactly the software it is going to be used by with your client.
- Have a report showing details of how the program ran as part of this testing. Try it on different operating systems and hardware.

DOCUMENT INTERNALLY AND EXTERNALLY

- Make a report to show the commenting on your code (internal documentation)
- Make a user guide to show the end user for how to actually use the software in their location

IMPLEMENT AND TEST WITH LIVE DATA

- Start the program in the client's work environment with actual data to see if it works.
- Do it over again on different computers to make sure it works for the client on all machines.

EVALUATE PERFORMANCE OF THE PROGRAM

- Survey the staff using the program to see if it works as expected
- Make note of what changes should be made and fix those
- Consider sending out an update of the software after you have again tested to make sure the changes have been successful

Congrats on finishing a topic! Keep going!!!

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Programming

CHARACTERISTICS OF SIMPLE DATA TYPES

Data Type	Range	Storage (Bytes)	Example
Integer	-2,147,483,648 to 2,147, 483,648	4	25,405
Real	Infinite numbers		2.4
Character	-	1	a’, ‘b’, ‘#’, ‘[‘
Boolean	-	2	True or false 1 or 0 On or off

SIMPLE DATA TYPES

- **Integer =** A whole number that can be positive, negative, or zero.
- **(Real) floating point number =** Numbers that contain floating decimal points. They can also be fractions.
- **Character =** A character is any letter, number, space, punctuation mark, or symbol that can be typed on a computer.
- **String =** A data type used to represent text rather than numbers. It's made up of characters that can also contain spaces and numbers.
- **Array =** A data structure that contains a group of elements. The elements are usually the same data type. Used to organise data so a related set of values can be sorted or searched easily.
- **Boolean =** Used for creating true/false statements. Uses the operators AND, XOR, OR, and NOT to compare values and return a true or false result.
  - **x AND y =** returns True if both x and y are true; returns False if either x or y are false.
  - **x OR y =** returns True if either x or y, or both x and y are true; returns False only if x and y are both false.
  - **x XOR y =** returns True if only x or y is true; returns False if x and y are both true or both false.
  - **NOT x =** returns True if x is false (or null); returns False if x is true.

CHARACTERISTICS OF COMPLEX DATA TYPES

<table>
<tr><th>Data Type</th><th>Explanation</th><th>Storage (Bytes)</th><th>Example</th></tr>
<tr><td>Record</td><td>A record refers to a row in a database. The range and size of the record are completely dependent on the content of said row.</td><td>-</td><td>Each individual horizontal row in this database is known as a record.
<table><tr><th>Id</th><th>Name</th><th>Email</th><th>Investments</th></tr><tr><td>231</td><td>Albert Master</td><td>albert.master@gmail.com</td><td>Bonds</td></tr><tr><td>210</td><td>Alfred Alan</td><td>aalan@gmail.com</td><td>Stocks</td></tr><tr><td>256</td><td>Alison Smart</td><td>asmart@biztalk.com</td><td>Residential Property</td></tr></table></td></tr>
<tr><td>1D Arrays</td><td>Also known as a single-dimension array, 1D arrays are used to hold a group of the same type (e.g. data type) items. Arrays are used heavily throughout programming of all types to store related information.</td><td>-</td><td># Example written in PHP

$classMates = array(
"William",
"Bob",
"Jessica",
"Freddie",
"Anne"
);
This provides an example of an array named classMates with 5 entries. All items within the array have the data type String.</td></tr>
<tr><td>String</td><td>-</td><td>Dependant on string size</td><td>‘This is a string of text’</td></tr>
</table>

PROGRAMMING CONCEPTS

CONSTANT

A known item or symbol that never changes its initial value restriction of normal operation. Useful for simplifying source code. Only needs to be changed in one place to take effect over the whole program.

VARIABLE

- A named value or symbol that can be changed as the program runs.
- **Declaring a variable =** To define the variable in terms of its name and data type. EX: int my_variable
- **Initialise a variable =** To give a variable a starting value.
  - Initialising a variable helps define its data type and undefined variables make debugging difficult.
- There are three different types of variables:
  - **Local =** A variable that's only accessible within the function that it's written in.
  - **Global =** A variable that's accessible from anywhere within the program
  - Parameter = A referenced variable or constant that's passed into a function or module. Usually have brackets around them so they are easy to find.

Appropriate Naming Conventions for Variables

- Variable name must be in the form $name or ${name}
- String name can consist of letters, numbers (0-9), and underscores
- The first character in the name must be a letter
- If ${name} form is used, you can also use spaces and don’t have to use underscores

Control Structures

Overview

- **Control structure:** Block of code that analyses variables and decides where to take the program
- Three types: Sequence, selection, and iteration
- Structure type being used depends on the type of task being carried out

Sequence

- Easiest structure to use and understand
- Runs through code in logical order

Selection

- Lets the program test conditions based off specific guidelines
- Follows IF [this] THEN [that] logic
- Can be seen as a boolean condition because the result is either true or false.
- Also relies on an ELSE statement.
- When specific guidelines aren't met the program follows ELSE instructions.

Iteration

- Another term for looping.
- Functions and programs that use looping will run a specific block of code until the criteria is met.
- Three types: While, do-while, and for.

While loops (test first)

- Can be seen as a repeating IF statement.
- At the statement start a boolean condition is given.
- If the while statement is true, the code within the while loop will execute.
- Once the statement is false the compiler or interpreter will move onto the next block of code.

For loops (fixed)

- Runs until a particular condition is satisfied.
- A boolean instruction is given at the start of the control structure.
- There's an incremental counter in it that tells it when to stop.

Do while loops (test last)

- Will run a condition regardless of previous outcomes.
    - This is because the boolean expression is evaluated at the end of the control structure.
        - Therefore it is guaranteed to run at least once

Stubs

A piece of code that acts as a place holder for code that's handled remotely or hasn't been developed yet. It can act like the code that's supposed to be there instead but with little functionality and smaller in size.

STATEMENTS

- A line or chunk of text written in a program just for documentation.
- Not processed when the program is executed.
- Documenting parts of code to help developers and end users use the software.

MODULARISATION

- Breaking up parts of code into modules to organise and lower the program's complexity.
- Similar to functions but don't have to return a value.
  - Modules are used to define task sets for the program to complete when asked to.

FUNCTIONS

-A procedure or routine a computer follows when executing a program.
-Sometimes defined as something that returns a value as opposed to a routine which doesn’t
-Allows us to reuse code without needing to rewrite it in different locations.
-Doesn't execute immediately after the program has loaded but when it's called upon.
-By breaking down programs into functions we can keep the code efficient and clean.

PARAMETER PASSING

Overview

- **Parameter passing =** The technique of passing values between functions or subroutines
- 2 common methods: value and reference passing
- **Parameter scope =** The boundary of which a value can exist
- Identifier lifetime is dependant on the type of passing being performed

Value Passing

- A parameter called by value outputs the value of the actual parameter.
- It's essentially passing a copy of the original value to the function or submodule.
- If the value of the original parameter changes the copy does not reflect the change.

Reference Passing

- A parameter passed by reference refers to the original location of the parameter's value.
- Changes to the original value will be reflected in our own value because our value is just a link to the original one.

DIFFERENCE BETWEEN SOURCE CODE, BYTE CODE, AND EXECUTABLE CODE

Source Code	Byte Code	Executable Code
- Can be seen as the 'before' code of a program. - Exists as plain text that's yet to be compiled or interpreted. - Contains the exact content of the program to be complied yet it doesn't perform any function. - Once the code is saved as a file it becomes source code until it is executed.	- Code processed by a virtual machine. - The virtual machine translates it to machine code.	- Low level language thats executed after being compiled or interpreted as source cod. - Unlike source code it is responsible or performing functions. - Hardware specific.

DIFFERENCE BETWEEN AN INTERPRETER & A COMPILER

Interpreter	Compiler
- Analyses and executes a program line by line. - Runs each line sequentially. - Overall execution time is slower. - Great for debugging since it executes lines until an error appears.	- Translates and executes a program as a whole. - First translates the entire program into machine code. - Generally faster than an interpreter because the code is complied in one block. - Uses less memory than interpreters. - Error messages only occur after the whole program is analysed

TYPES OF PROGRAM OR CODE ERRORS

SYNTAX ERRORS

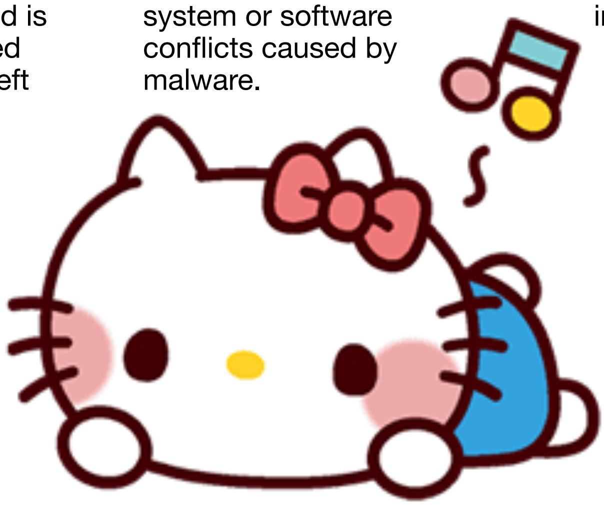
- A programming mistake that breaks a program's rules or conventions.
- Can occur as small mistakes and is often hard to pick up without running the program.
- Occur when a command is mistyped or an expected phrase or argument is left out.

RUN-TIME ERRORS

- Software and hardware faults that stop a program from running.
- The software itself doesn't have an internal error.
- Caused by physical damage to the computer system or software conflicts caused by malware.

LOGICAL ERRORS

- Occurs when the program gives wrong or unexpected results because of poor logical processing.
- Caused by a programmer's mistakes inside the source code.



PURPOSE AND CHARACTERISTICS OF INTERNAL AND EXTERNAL DOCUMENTATION

INTERNAL DOCUMENTATION

- Internal documentation = documentation written within a program.
  - Refers to comments and instructions written by and for the developers.
- Used to keep a form of communication open between developers.
- Used to explain parts of code as well as provide useful information on the program internally.

EXTERNAL DOCUMENTATION

- External documentation = manuals and guides written for end users.
- They explain how to operate a program and use it as intended.
- They are written by developers and other end users.

TYPES OF DATA VALIDATION TECHNIQUES

RANGE CHECKING

- Used to make sure that our data falls within the boundaries that we set.
- We can look at the scope of values that our data can fall within.
- Mostly used on data involving numbers and figures.

Type	Description	Data Validation
Maximum	The maximum grade possible is 100%	$X \leq 100$
Minimum	The minimum grade possible is 0%	$X \geq 0$
Range	To achieve an ‘A’ you must achieve between 75% and 100%	$75 \leq X \leq 100$

TYPE CHECKING

- A simple check on the type of data that's being put into or pullout out from a function.
- We do this to ensure that the program won't break from an incompatible data type.

Type	Description	Data Validation
Number	The number entered must be in numerical form. EX: not ‘ten’	10 ≠ ten
Email	The email address must follow correct syntax and resolve to a real domain	email@domain.tld
Date	The date must be valid. EX: There is no 32nd day or 13th month	32/13/2000

Congrats on finishing a topic! Keep going!!!

Networks and Communications

Roles of Hardware Devices in Network Communications

Router

A router is a device that forwards data packets along networks.
A router is connected to at least two networks, commonly two LANs or WANs or a LAN and its ISP's network.
Routers are located at gateways, the places where two or more networks connect.

Switch

A network switch (also called switching hub, bridging hub, officially MAC bridge) is a computer networking device that connects devices together on a computer network by using packet switching to receive, process, and forward data to the destination device.

Firewall

A firewall is a barrier to control crossover of network traffic both in and out of Internet-connected network, or between different segments of an internal network.
Firewalls are used to protect both home and corporate networks.
A typical firewall program or hardware device filters all information coming through the Internet to your network or computer system.

Modem

Short for modulator-demodulator.
A modem is a device or program that enables a computer to transmit data over, for example, telephone or cable lines.
Computer information is stored digitally, whereas information transmitted over telephone lines is transmitted in the form of analogue waves.

Network Interface Card (NIC)

To connect to a network, a computer uses a network interface card (NIC). A NIC controls the wired and wireless connections of a computer to exchange information with other computers and the Internet.

Wireless Access Point

In computer networking, a wireless access point (WAP), or more generally just access point (AP), is a networking hardware device that allows a Wi-Fi device to connect to a wired network.

BRIDGE

A device that governs the flow of traffic between networks or network segments and transfers packets between them.

GATEWAY

In enterprises, the gateway is the computer that routes the traffic from a workstation to the outside network that is serving the Web pages.
In homes, the gateway is the ISP that connects the user to the Internet.
In enterprises, the gateway node often acts as a proxy server and a firewall.

REPEATERS

Network repeaters regenerate incoming electrical, wireless or optical signals.
Repeaters attempt to preserve signal integrity and extend the distance over which data can safely travel.
Actual network devices that serve as repeaters usually have some other name.

Role of Layers within Department of Defence TCP/IP Model of in Network Communications:
The Department of Defence created TCP/IP to ensure and preserve date integrity. The (DoD) model is a condensed version of the OSI model and only has four layers.

PURPOSE OF LAYERS WITHIN DOD TCP/IP MODEL

APPLICATION LAYER

Defines protocols for node-to-node application communication and also controls user interface specifications. Consists of a set of services that provide ubiquitous access to all types of networks. Applications utilise the services to communicate with other devices and remote applications

TRANSPORT LAYER

This layer shields the upper layers from the process of sending data. Also provides an end-to-end connection between two devices during communication by performing sequencing, acknowledgments, checksums, and flow control. Applications using services at this layer can use two different protocols: TCP and UDP.

INTERNET LAYER

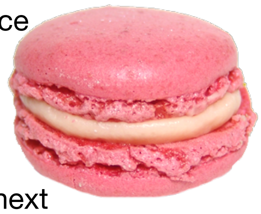
The Internet layer exists for routing and providing a single network interface to the upper layers. IP provides the single network interface for the upper layers.

Network Layer
The Network layer monitors the data exchange between the host and the network. Oversees MAC addressing and defines protocols for the physical transmission of data.

Characteristics of Wireless Transmission Media

Broadcast Radio
- Effective for broadcast communications
- Transmission is limited to line of signs and distant transmitters won't interfere with each other due to atmosphere reflection
- Less sensitive to weakening from rainfall
- Multi-path interference imparts broadcast radio waves
  - Reflection of waves from land, water, and natural or human-made objects can create multiple paths between antennas

Satellite
- Gets radio wave signal from ground station on Earth, then bounces signal back down to a different ground station
- Satellite also acts like a repeater by amplifying the signal so that it can travel further
  - Means that the radio wave can go a much further distance than just 'line of sight' on the ground
- Can't transmit and receive signals on the same frequency without interference



Microwave
- Commonly used for telephone systems
- Has a series of large towers following a path from one town/location to the next
- Signal goes from one tower, is amplified, and sent on to the next tower
- Its 'line of signs' and can't go through mountains or major forests

Cellular
- Works on a network or collection of cells, each having base station and mast
- Phone calls travel back and forth along these cells
- Don't work in most remote areas because there are no available cells

Characteristics of Wired Transmission Media

Twisted Pair
Overview
- Used to transmit both digital and analog transmission
- For analog signals amplifiers are required every 5-6km
- For digital signals amplifiers are required ever 2-3km
- Limited in distance, signals, bandwidth, and data rates
- Susceptible to interference and noise because of its easy coupling with electromagnetic fields
- Shielding reduces interference

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Unshielded Twisted Pair

- Ordinary telephone wire
- Least expensive type of transmission media commonly use for LANs
- Easy to work with and install
- Subject to external electromagnetic interference

Shielded Twisted Pair

- Provides better performance at higher data rates
- More expensive and difficult to work with than unshielded twisted pair

FIRE OPTICS

- In optical fibre technology, single mode fibre is optical fibre that is designed for the transmission of a single ray or mode of light as a carrier and is used for long-distance signal transmission.
- For short distances, multimode fibre is used. In optical fibre technology, multimode fibre is optical fibre that is designed to carry multiple light rays or modes concurrently, each at a slightly different reflection angle within the optical fibre core.
- For longer distances, single mode fibre is used.

CSMA/CD VS CSMA/CA

- **CSMA/CD =** Carrier sense multiple access with carrier detection
- **CSMA/CA =** Carrier sense multiple access with carrier avoidance

CSMA/CA	Similarities	CSMA/CD
- CSMA/CA is for wireless networks. - CSMA/CA uses the Wireless Access Point to control who sends and when they are clear to send. - Only transmit when channel is idle - Transmit packet data in its entirety	- Both use a system of sending and receiving data packets (information) - Both use Carrier Sensing (there is there a carrier) - Both use Multiple Access (others users around)	- CSMA/CD is for wired or ethernet networks. - CSMA/CD it provides a listen, then transmit system - CSMA/CD, if there is a collision, it waits a random time 0.5-10 secs, then sends again. - CSMA/CD, each computer waits a different random time, then sends again.

ERROR DETECTION AND CORRECTION

PARITY BIT

- Parity bit = a single bit added to a string of binary code so that the total number of 1 bits in the string is even or odd
- In parity check the sender and receiver decide whether an odd or even will be required or error detection when handshaking
- Odd parity = a parity with an odd amount of ones when a parity bit is used
- Even parity = a parity with an even amount of ones when a parity bit is used
- Error detected when the parity isn't at the receiving end is different from the sending end
- Only detects errors relating to parity so other errors aren't noticed
- Adding an extra bit to every byte will have significant increase on the amount of data being transmitted

Data	Parity Bit	Data with Parity	Amount of ones	Odd/Even Parity
1	1	11	2	Even
1010	1	10101	3	Odd

CHECKSUM

Overview

- This uses a calculation to check if all the bits are added up correctly.
- If when summed up, doesn't match the number added to the right hand end of the number (the checksum), it must be wrong and the message must be sent again.

How it Works

- The message or image is converted to bits or numbers.
- The numbers are added up.
- The previously agreed to number, is divided into the total of the numbers.
- The remainder or modulus, is added to the right hand end of the number to be sent, and then sent with the message.
- If the message is received and upon checking it notices that the checksum on the right hand side, does not match the calculation, it asks for a resend.

TYPES OF COMMUNICATIONS PROTOCOLS AND STANDARDS

WIRELESS

Bluetooth

- Standard for exchanging data over short distances from fixed and mobile devices, and building personal area networks
- Invented in 1994 by telecom vender Ericsson
- Managed by Bluetooth Special Interest Group
- Uses radio communications system, therefore devices don't have to be in visual line of sight of each other
  - Instead a quasi optical wireless path must be possible

Ethernet 802.11x

- IEEE802.11 is standard for defining communication over a wireless LAN (WLAN)
- Provides 1 or 2 Mbps transmission in the 2.4 GHz band using either frequency hopping spread spectrum (FHSS) or direct sequence spread spectrum (DSSS).
- Specifies over-the-air interface between a wireless client and a base station or between to wireless clients

Radio Frequency Identification (RFID)

- Uses electromagnetic fields to automatically identify and track tags attached to objects
  - Tags contain electronically stored information
- Unlike a barcode, the page doesn't need to be in the line of sight of the reader so it can be placed in the tracked object
- A method for Automatic Identification and Data Capture (AIDC)
- Raises privacy concerns because of the possibility of reading personally-linked information without consent

Wireless Application Protocol (WAP)

- Technical standaard for accessing information over a mobile wireless network
- Standardises the way that wireless devices can be used for internet access can be used for internet access
- Its layers are wireless application environment (WAE), wireless session protocol (WSP), wireless transaction protocol (WTP), wireless transport layer security (WTLS), and wireless datagram protocol (WDP)

WIRED

Ethernet 802.3

- Standard specification for ethernet (method of physical communication is a local area network) maintained by the Institute of Electrical and Electronics Engineers (IEEE)
- In general it specifies physicals medial and working characteristics of ethernet
- 802.3 also define LAN access method using CSMA/CD.

TCP/IP

IP4

- Internet Protocol version 4 (IPv4) is the fourth version of the Internet Protocol (IP).
- It is one of the core protocols of standards-based inter-networking methods in the Internet, and was the first version deployed for production in the ARPANET in 1983.
- It still routes most Internet traffic today,[1] despite the ongoing deployment of a successor protocol, IPv6.
- IPv4 is a connectionless protocol for use on packet-switched networks.
- It operates on a best effort delivery model, in that it does not guarantee delivery, nor does it assure proper sequencing or avoidance of duplicate delivery.
- These aspects, including data integrity, are addressed by an upper layer transport protocol, such as the Transmission Control Protocol (TCP).

IP6

- Internet Protocol version 6 (IPv6) is the most recent version of the Internet Protocol (IP), the communications protocol that provides an identification and location system for computers on networks and routes traffic across the Internet. I
- Pv6 was developed by the Internet Engineering Task Force (IETF) to deal with the long-anticipated problem of IPv4 address exhaustion.

Dynamic Host Configuration Protocol (DHCP)

- Client/server protocol that automatically provides an internet protocol host with its IP address and other related configuration information
- Without DHCP, IP addresses for new computers or computers that are moved from one subnet to another must be configured manually
  - With DHCP this process is automated and managed centrally
- DHCP server maintains IP address pool and leases address to any DHCP-enabled client when it is requested

Domain Name Server (DNS)

- Maintains a directory of domain names and translates them into IP addresses
- After registering a new domain name or update DNS servers on your domain name, it takes around 12-36 hours for DNS to be updated and able to access the information
  - The 12-36 hour period is known as **propagation**

SAN and NAS

Storage Area Network (SAN)

- Dedicated high speed network (or subnetwork) that interconnects and presents shared pools of storage devices to multiple servers
- Primarily used for enhancing storage devices that are accessible to servers so the devices seem like locally attached devices to the operating system

Network Attached Storage (NAS)

- File level computer data storage server connected to a computer network providing data access to an outside group of clients
- A networked appliance that contains one or more storage drives that are often arranged in RAID

SAN	Similarities	NAS
- SAN is like a large-hard-disk shared among different users. If a user/server wants to access storage, it will be given a dedicated portion of the total available space (called a LUN). The server has to manage files itself and the SAN unit only understands **block-level addresses.** - SAN is generally faster and the advantage is having the **hardware shared** across servers to avoid wasted capacity. Each LUN is private to the server and can grow or shrink in capacity based on requirement.	- The various computers in a network, such as individual users' desktop computers and dedicated servers running applications ("application servers"), can share a more centralised collection of storage devices via a network connection such as a local area network (LAN). - Concentrating storage in NAS servers or a SAN, instead of placing storage devices on each application server, allows application server configurations to be optimised for running their applications instead of also storing all the related data and moves the storage management task to the NAS or SAN system. Can reduce the amount of excess storage that must be purchased and provisioned as spare space.	- NAS is like a another computer on your network which you can ask for so-and-so files. It gives you the file directly and internally manages how data is stored, how files are backed up etc etc. NAS unit understands **files, folders and volumes**. - NAS is great for **sharing files** across users and servers. Both generally offer features like snapshot and backup as well.

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NETWORK SECURITY METHODS

FIREWALLS

- A firewall is a network security system, either hardware- or software-based, that uses rules to control incoming and outgoing network traffic.
- A firewall acts as a barrier between a trusted network and an untrusted network. A firewall controls access to the resources of a network through a positive control model. This means that the only traffic allowed onto the network is defined in the firewall policy; all other traffic is denied.

ANTI-VIRUS SOFTWARE

- Antivirus (or anti-virus) software is used to safeguard a computer from malware, including viruses, computer worms, and Trojan horses.
- Antivirus software may also remove or prevent spyware and adware, along with other forms of malicious programs.

PASSWORD AND NETWORK USER POLICIES

A password is a word or string of characters used for user authentication to prove identity or access approval to gain access to a resource (example: an access code is a type of password), which is to be kept secret from those not allowed access.

AUTHENTICATION

- Authentication is the process of determining whether someone or something is, in fact, who or what it is declared to be.
- Logically, authentication precedes authorization (although they may often seem to be combined).
- The two terms are often used synonymously but they are two different processes.

ENCRYPTION

In cryptography, encryption is the process of encoding a message or information in such a way that only authorised parties can access it.
Encryption does not itself prevent interference, but denies the intelligible content to a would-be interceptor.
In an encryption scheme, the intended information or message, referred to as plaintext, is encrypted using an encryption algorithm – a cipher – generating cipher text that can only be read if decrypted.
For technical reasons, an encryption scheme usually uses a pseudo-random encryption key generated by an algorithm.
It is in principle possible to decrypt the message without possessing the key, but, for a well-designed encryption scheme, considerable computational resources and skills are required.
An authorised recipient can easily decrypt the message with the key provided by the originator to recipients but not to unauthorised users.

Compromising Network Security

Denial of Service

- An attempt to make a machine or network resource unavailable to its intended users, such as to temporarily or indefinitely interrupt or suspend services of a host connected to the Internet.
- **Distributed denial-of-service (DDoS):** When the attack source is more than one, often thousands of, unique IP addresses. It is analogous to a group of people crowding the entry door or gate to a shop or business, and not letting legitimate parties enter into the shop or business, disrupting normal operations.
- Criminal perpetrators of DoS attacks often target sites or services hosted on high-profile web servers such as banks, credit card payment gateways.

Backdoors

- A method, often secret, of bypassing normal authentication in a product, computer system, crypto-system, or algorithm
- Backdoors are often used for securing unauthorised remote access to a computer, or obtaining access to plaintext in cryptographic systems.
- A backdoor may take the form of a hidden part of a program, a separate program (e.g. Back Orifice may subvert the system through a rootkit), or may be a hardware feature.
- Although normally surreptitiously installed, in some cases backdoors are deliberate and widely known.
  - These kinds of backdoors might have "legitimate" uses EX: providing the manufacturer with a way to restore user passwords.

IP Spoofing

- The creation of Internet Protocol (IP) packets with a forged source IP address, with the purpose of hiding the identity of the sender or impersonating another computing system.
- By forging the header so it contains a different address, an attacker can make it appear that the packet was sent by a different machine.
- Mainly used when the attacker does not care about the response or the attacker has some way of guessing the response.

Phishing

- Attempt to acquire sensitive information, often for malicious reasons, by masquerading as a trustworthy entity in an electronic communication.
- Typically carried out by email spoofing or instant messaging, and it often directs users to enter details at a fake website whose look and feel are almost identical to the legitimate one.

Network Performance Factors

Bandwidth

Every Internet connection has a specified maximum bandwidth, but many factors can combine to limit this for a particular device. This results in a slowness of the perceived speed of a connection. The factors that affect bandwidth can originate in an individual computer and the nature of the Internet connection itself.

Network Design:

The design of a network affects its performance. If excessive cabling is used, the speed may be slowed down.

DATA COLLISIONS

- A collision occurs on your network when something happens to the data sent from the physical network medium that prevents it from reaching its destination.
- Mainly, it encounters another signal from another host on the network that yields a resulting useless signal on the network when the signals combine.
- The collision occurs when the sending device does not receive a clear response back within the allotted time.
- This causes an issue for both network devices because they both need to wait for an ever-increasing period until they are able to transmit the data clearly.
- If the network is busy enough, the network devices can spend an inordinate amount of time retransmitting data.
- The network area where a collision may occur is called a collision domain.

EXCESS BROADCAST TRAFFIC

- Excessive broadcast traffic can sometimes create a broadcast storm.
- A broadcast storm occurs when messages are broadcast on a network and each message prompts a receiving node to respond by broadcasting its own messages on the network.
- This, in turn, prompts further responses that create a snowball effect.

Congratulations on reaching the end of (almost) all of the knowledge required for your final exam! Now go forth and do as many exams in preparation as you possibly can! Refer back to this document to refresh your knowledge when you see fit and may SCSA not screw you over (for this exam) this year!!